

Project Submission Report: Movie Ticket Booking Management

CineWave Entertainment Case Management System • Pega Platform '26

Application: Movie Ticket Booking Management

Case Type: Movie Ticket Request (Prefix: M-)

Platform Environment: Pega Infinity v26 (Pega Academy)

Submission Status:

COMPLETED • 100%

1. Executive Summary & Objectives

CineWave Entertainment previously handled ticket bookings and tracking manually across multiple theatres through disconnected email and offline systems. This project implemented an automated, end-to-end case lifecycle on the **Pega Platform** to capture customer requests, automate availability validation, compute booking pricing, enforce approval routing via dedicated work queues, configure service-level agreements (SLAs), and trigger automated email correspondence upon resolution.

2. Case Lifecycle Architecture

Primary Stages & Step Flow:

- Initial Stage (Event Setup):** Collects user data via *Submit Movie Ticket Request* form and calculates initial charges via *Calculate Booking Cost*.
- Availability (Inventory Management):** Evaluates theater seating inventory via *Check Show Availability*.
- Approval (Sales Monitoring):** Routes the review assignment (*Confirm Booking Request*) to specialized work queues based on show category, followed by *Review Booking Details*.
- Booking Execution:** Processes the ticket confirmation via Data Transform (*Process Ticket Booking*) and generates customer email notifications (*Notify Booking Confirmation*) resulting in **Resolved-Completed** status.
- Approval Rejection (Alternate Stage):** Handles rejected requests cleanly.

[Screenshot 1: Case Lifecycle View]

Attach screenshot of the Case Designer showing all 4 primary stages (Initial Stage, Availability, Approval, Booking Execution) and the Alternate Stage.

3. User Story Implementation Summary

Story ID	User Story Title	Implementation Method	Status
US-001	Submit Movie Ticket Request	Collect Information Form with customer contact and show fields	VERIFIED
US-002	Check Show Availability		VERIFIED

Story ID	User Story Title	Implementation Method	Status
US-003	Calculate Booking Cost	Inventory validation step & conditional path	
		Calculated property: TotalCost = (TicketQuantity * TicketPrice) - Discount	VERIFIED
US-004	Confirm Booking Request	Approval step with configurable routing logic	VERIFIED
US-005	Maintain Movie and Show Data	Data objects and picklists (Show Type, Movie Details)	VERIFIED
US-006	Review Booking Details	Read-only review view prior to final ticket commitment	VERIFIED
US-007	Process Ticket Booking	Data Transform generating unique confirmation number	VERIFIED
US-008	Notify Booking Confirmation	Automated Send Email step mapped to .CustomerEmail & case details	VERIFIED
US-009	Define Booking SLA	Case-level SLA configured with Goal: 1 Day and Deadline: 2 Days	VERIFIED
US-010	Route Request by Show Type	Business logic routing: Premium → PremiumShowQueue, Standard → StandardShowQueue	VERIFIED

4. Detailed Technical Configurations

A. Data Model & Key Properties

Property Name	Field Name	Type	UI Control / Behavior
Customer Name	pyCustomerName	Text	Single Line Text Input
Customer Email	CustomerEmail	Email	Validated Email Address
Show Type	ShowType	Picklist	Dropdown List (Standard, Premium, IMAX)
Payment Method	PaymentMethod	Picklist	Dropdown List (Credit Card, UPI Transfer)
Ticket Quantity	TicketQuantity	Integer	Numeric Input
Ticket Price	TicketPrice	Currency	Currency Formatted
Total Cost	TotalCost	Currency	Calculated Expression Field
Confirmation Number	ConfirmationNumber	Text	Generated Confirmation Code

[Screenshot 2: Data Model View]

Attach screenshot of the Data Model tab showing the configured properties.

B. SLA & Queue Routing Logic

- **SLA Configuration:** Case Service Level Agreement enabled under Settings → Goal & Deadline.
 - **Goal:** 1 Day (Urgency +10)
 - **Deadline:** 2 Days (Urgency +10)
- **Routing Rule Logic:** Step *Confirm Booking Request* uses Business Logic routing:
 - If `.ShowType == "Premium"` → Route to **PremiumShowQueue**
 - Otherwise → Route to **StandardShowQueue**

[Screenshot 3: SLA & Routing Setup]

Attach screenshot showing the Goal & Deadline settings and the Business Condition routing popup.

5. End-to-End Execution Results

Test Case Execution Verification:

- **Case Reference:** Case M-2016 / M-2017
- **Final Status:** **RESOLVED-COMPLETED**
- **Attachment Verification:** Correspondence attachment Movie Ticket Confirmation automatically generated and attached to the case history.

[Screenshot 4: Resolved Case & Email Correspondence]

Attach screenshot of the final case view showing Resolved-Completed status and the email correspondence item under Attachments.

6. Submission Links & Deliverables

- **GitHub Repository:** Public repository containing application artifacts and documentation.
- **Video Demo Link:** Google Drive public recording showing case creation, approval, and email resolution.